National University of Computer and Emerging Sciences, Lahore Campus

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| Course Name: | Programming Fundamentals | Course Code: | CS-1002 |
| Degree Program: | BS(CS-SE-DS) | Semester: | Fall 2021 |
| Exam Duration: | 60 Minutes | Total Marks: | 30 |
| Paper Date: | 21-OCT-2021 | Weight | |
| Section: | ALL | Page(s): | 4 |
| Exam Type: | Midterm-I | | |

**Student : Name:_____ Roll No._____ Section:_____**

Instruction/Notes: Calculators are not allowed. Answer in the space provided. Show all working. Use last page for rough work. **ROUGH SHEETS NOT ALLOWED**. Good luck!

**Question1.** Write a C++ program OR pseudocode for this problem. Take 3 numbers as input from the user and print the median number while strictly following the output format. When listing the 3 numbers in ascending order, the number that comes in the middle is the median number. Consider some sample examples. Note, these examples are just for explanation purposes, your code must work for any input. Properly indent your code. (10 Points)

| | |
|---|---|
| **Input**: 3 44 21<br>**Output**: The median of 3 44 and 21 is 21<br><br>**Input**: 3 44 3<br>**Output**: The median of 3 44 and 3 is 3 | **Input**: 17 90 11<br>**Output**: The median of 17 90 11 is 17<br><br>**Input**: 1 5 18<br>**Output**: The median of 1 5 and 18 is 5 |

Roll Number:_____ Section: _____

**Question 2.** Dry run the given C++ code and answer the questions given below. (10 Points)

```
#include <iostream>
using namespace std;
int main()
{
     int result = 0;
     int num1 = 0;
     int num2 = 10;
     int count = 20;
     while( count >= 8)
     {
          if(num1 != 0)
          {
               if( num2 <= num1)
                    result = num1 % num2;
               else
                    result = num2 % num1;
          }
          else
               result = num2;
          count = count – num1;
          num1 = num1 + 2;
          num2 = num2 – 1;
     }
     cout<< num1 << “,  ”  <<num2 << “,  ” <<result << “,  ” <<count <<endl;
     return 0;
}
```

a. Fill the table according to values of variables **after each iteration:**

| Iteration # | result | count | num1 | num2 |
|---|---|---|---|---|
| **0** | **0** | **20** | **0** | **10** |
| 1 | 10 | 20 | 2 | 9 |
| 2 | 1 | 10 | 4 | 8 |
| 3 | 0 | 14 | 6 | 7 |
| 4 | 1 | 8 | 8 | 6 |
| | | | | |
| | | | | |
| | | | | |

b. What is the output of the program?

10, 5, 2, 0

Roll Number:_____ Section: _____

**Question3.** (5+5 Points)

a. Consider the following program fragment:

```
int i = 4;
int j = 17;
float z = 2.6;
```

Determine the value of each expression below. If the result is a floating-point value, include a decimal point in your answer.

| | Expression | Expected value |
|---|---|---|
| **a.** | `i/j` | |
| **c.** | `z*j` | |
| **e.** | `(1/2)*i` | |
| **g.** | `j/2` | |
| **i.** | `i+j/z` | |

| | Expression | Expected value |
|---|---|---|
| **b.** | `1.0/i+2` | |
| **d.** | `i+j%i` | |
| **f.** | `2*i+j-i` | |
| **h.** | `2*3-1%3` | |
| **j.** | `(z>0.5 && j==17)` | |

b. Identify and correct all syntax errors in the following program. Encircle the error and provide corrections in the box right next to it.

| | Specify corrections for respective lines here! |
|---|---|
| `#include <iostream>;` | |
| `using namespace std;` | |
| `int Main( )` | |
| `{` | |
| `int var1;` | |
| `Int var2, var3=5;` | |
| `var1=12` | |
| `int square;` | |
| `var1 * var1 = square;` | |
| `COUT << Square;` | |
| `cout >> end;` | |
| `return 0;` | |
| `}` | |

Roll Number:_____ Section: _____

//For rough work